

Worksheet 2

Stern-Gerlach and ProbabilitiesExploring the Stern-Gerlach Experiment • Normalization • Probabilities

1. This question uses the SPINS simulation which you can find online at <https://physics.weber.edu/schroeder/software/Spins.html>. Set up the experiment so that you have an oven, a Stern-Gerlach S_z device, and two counters, one for each output. Click on the number 1 on the oven; this will set each particle coming out to be in the initial state

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{i}{\sqrt{2}}|-\rangle.$$

- (a) First, predict what the experiment will measure: what are the probabilities $\mathcal{P}_+$ and $\mathcal{P}_-$ for measuring the particle to be spin up or spin down, respectively? Show your calculation.
- (b) Okay, now go ahead and click the 10k button to send in 10,000 particles. Are your predictions correct? Are they *perfectly* correct?
- (c) Next, replace the S_z device with one oriented along the x direction. Once again, make a prediction: what are the probabilities for measuring $\mathcal{P}_{+x}$ and $\mathcal{P}_{-x}$? Test your prediction once you've done the calculation.

- (d) Finally, replace the S_x device with one oriented along the y direction. What are the probabilities for measuring $\mathcal{P}_{+y}$ and $\mathcal{P}_{-y}$? Test your predictions. Any observations to make here?

2. Now for something a little trickier. One the SPINS experiment, keep the single Stern-Gerlach device (any orientation is fine), but click on the number 3. This one sends out particles all in the same state $|\psi_3\rangle$, but we don't know exactly what that state is. Our job is to figure it out.

- (a) First, run your experiments: using all three orientations for the SG device, figure out the probabilities for spin up and spin down for each and record them below.

Probabilities	S_x	S_y	S_z
Spin up			
Spin down			

- (b) Write the unknown state $|\psi_3\rangle$ as

$$|\psi_3\rangle = a|+\rangle + b|-\rangle,$$

where a and b are unknown *complex* constants. This state is written in the S_z basis, so start with your results for that orientation in table. Calculate $\mathcal{P}_+$ and $\mathcal{P}_-$ and use those results to find out what $|a|$ and $|b|$ are.

- (c) Unfortunately, we only know the magnitudes of those complex numbers a and b . Since we know (or will soon) that an overall phase has no physical meaning, we can write the state now as

$$|\psi_3\rangle = |a||+\rangle + |b|e^{-i\phi}|-\rangle,$$

where ϕ is the phase – our last unknown. Calculate $\mathcal{P}_{+x}$ and $\mathcal{P}_{-x}$ and use your results from the S_x experiment to find what ϕ is (careful – there are multiple possibilities it could be!). *Hint:* You might need to use the relationship

$$\cos \phi = \frac{e^{i\phi} + e^{-i\phi}}{2}.$$

- (d) Okay, one last problem – we don't know the sign of the phase ϕ (it could be positive or negative). To resolve this last ambiguity, we need to use our results for S_y – go ahead, and write the final ket of the state $|\psi_3\rangle$.